

Prajwal Umesh Nagalakshmi

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Technical Skills

Languages: Python, Java, TypeScript/JavaScript

Systems: Unix/Linux, Git

Back end & Distributed: Spring Boot, Django, FastAPI, gRPC & Protocol Buffers, Microservices, Async/Concurrency, SAGA, REST APIs

Front end & Mobile: React, Next.js, React Native, Tailwind CSS

AI & Security: OpenAI API (GPT-4o, Whisper), agent/security monitoring, Cursor, Vercel

Database: PostgreSQL (RDS), MySQL, MongoDB, Prisma ORM

Cloud & DevOps: AWS (EC2, S3, RDS), Docker & Docker Compose, CI/CD (GitHub Actions), Firebase

Professional Experience

Software Engineer Intern | Appleute | Bangalore, India

Oct 2021 - Mar 2022

- Developed **Betra** - AI-powered cashierless shopping app in retail stores.
- Improved **user engagement** by **40%** through **React Native** performance optimization.
- Reduced API response times by **25%** with backend **caching** and **NestJS** optimization.
- Integrated **Internationalization** and **local storage** to enable **multi-language** and **offline support**.

Projects

Agent Sentinel - Monitoring SDK | Python, FastAPI, Next.js | [Github](#)

Jul 2025 - Aug 2025

- Built enterprise **AI Agents** security monitoring **SDK** with real-time threat detection using a decorator.
- Developed a 5-stage **multi-agent** intelligence engine for automated security analysis and reporting.
- Built a **FastAPI** backend with threat detection algorithms and **Next.js/TypeScript** dashboard for real-time security monitoring.
- Wired **distributed tracing** (Weave) to follow requests across services and surface anomalies.

DealCart+ | Spring Boot, gRPC/Protocol Buffers, Docker, AWS | [Github](#)

Sept 2025 - Sept 2025

- Architected Java 21 **microservices** (**ProtoBuf/gRPC**, Spring Boot, React/Tailwind) with SSE streaming.
- Implemented parallel vendor fan-out, server-streamed quotes, token-bucket **rate limiting** + **request tracing**.
- Designed **promise-graph** checkout with timeouts/retries, **idempotency**, **SAGA** rollbacks.
- Built production-style **auto-scaling**: CPU/memory-based scaling (1→5 instances) with 5-10min cooldowns.
- Developed system metrics API: CPU, memory, load average, latency, error rate monitoring.
- Load tested with simulated traffic: **200K+ requests** in high-concurrency bursts, **100% success rate**, P95 ~250ms, auto-scaling validated.
- Deployed on **AWS EC2** (Docker Compose) + S3 receipts + GitHub Actions **CI/CD** → GHCR.

ReKindle | React Native, Expo, OpenAI GPT-4o, Whisper API | [Github](#)

Jun 2025 - Jun 2025

- Built a networking **mobile app** with card/LinkedIn **QR scan**, **transcription** and **AI follow-up generation**.
- Implemented GPT-4 personalized messages based on **conversation context**.
- Designed a privacy-first **local storage** with offline support and intelligent search.
- Added dark mode, contact history, and message preview/edit functionality.

Achievements

AgentBay | AI Agents + MCP Hackathon | 3rd place | [Github](#)

Aug 2025 - Aug 2025

- Developed a **multi-agent** conversational platform with agents for YouTube, finance, and travel queries.
- Implemented **context-aware** agent coordination using **Model Context Protocol (MCP)** standard.

Education

California State University, Dominguez Hill

Aug 2023 – Dec 2025

Master of Science in Computer Science, GPA: 3.72

California, US

Coursework: Data Structures, Algorithm Analysis, Object Oriented Analysis and Design, Data Science, Adv Software Engineering, Adv Programming, Software Project, Adv Operating System, Security Engineering

RNS Institute of Technology

Jun 2021

Bachelor of Engineering in Information Science Engineering

Bangalore, India

Coursework: DBMS, Python, Programming in C, Web Technology, Operating Systems(Unix/Linux)